Yongchao Ma

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Education

2022–2026 **University of Michigan, Ann Arbor**, USA
PH.D. in Survey and Data Science
Dissertation: *On effective and efficient survey data collection at the crossroads of statistics and AI*
Committee: James Wagner (Chair), Brady T. West, Michael Traugott, Barry Schouten

2019–2021 **Utrecht University**, The Netherlands
M.Sc. in Methodology and Statistics, *cum laude*
Track: European Master in Official Statistics
Thesis: *Optimal stratification in Bayesian adaptive survey design*
Supervisors: Nino Mushkudiani, Barry Schouten

2018–2019 **Linköping University**, Sweden
One-term graduate exchange, Computational Social Science

2017–2019 **University of Macau**, Macao SAR
M.S.Sc. in Criminology and Criminal Justice

Research Appointments and Experience

2022–2025 (0.5 FTE) Graduate Student Research Assistant, **University of Michigan, Ann Arbor**, USA

- Designed a daily web survey of time-varying cannabis use motives and analyzed temporal patterns using random-intercept latent transition analysis.
- Analyzed survey experiments on motivated reasoning and poll credibility using structural equation modeling; investigated cross-national measurement invariance of survey attitudes.

2021–2022 (1.0 FTE) Research Engineer, **Utrecht University**, The Netherlands

- Developed ASReview LAB interfaces in React and REST APIs connecting the interface to Python servers; contributed to the first stable release.
- Wrote developer and user documentation and improved usability through feedback on GitHub; continued as a volunteer maintainer after the appointment.

2020–2021 (0.4 FTE) Graduate Student Research Assistant, **Utrecht University**, The Netherlands

- Processed metadata for systematic review datasets in Python and developed word-cloud visualizations of titles and abstracts.

2020–2021 (1.0 FTE) Intern Survey Methodologist, **Statistics Netherlands**, The Netherlands

- Developed and implemented optimization methods in R for Bayesian adaptive survey design, addressing nonresponse bias under budget constraints.

Grants & Fellowships

Winter 2026 Rackham One-Term Dissertation Fellowship, $14,596 + Tuition & Fees
Rackham Graduate School, University of Michigan

2024–2025 Charles Cannell Fund in Survey Methodology, $10,000
Institute for Social Research Next Generation Initiative, University of Michigan
Project: *Leveraging survey variables to estimate response propensity in Bayesian adaptive survey design: A simulation study*

Honors & Awards

2025 Rackham Conference Travel Funding, Rackham Graduate School, University of Michigan
2024 Rackham Conference Travel Funding, Rackham Graduate School, University of Michigan
2023 Student Travel Grant, Program in Survey and Data Science, University of Michigan
2023 Student Fellowship, Midwest Association for Public Opinion Research
2018 Outgoing Exchange Scholarship, University of Macau

Publications

Refereed journal articles

2026 **Ma**, **Y.**, Mushkudiani, N., & Schouten, B. (2026). An optimal stratification method for addressing nonresponse bias in Bayesian adaptive survey design. *Sociological Methods & Research*, *55*(3), 1064–1092. https://doi.org/10.1177/00491241251345463

2026 **Ma**, **Y.**, West, B. T., & McCabe, S. E. (2026). Daily web survey data collection of time-varying cannabis use motives and contexts, with implications for adaptive interventions: A pilot study. *Drug and Alcohol Dependence*, *278*. https://doi.org/10.1016/j.drugalcdep.2025.112974

2024 West, B. T., **Ma**, **Y.**, Lankenau, S., Wong, C. F., Bonar, E. E., Patrick, M. E., Walton, M. A., & McCabe, S. E. (2024). Latent transition analysis of time-varying cannabis use motives to inform adaptive interventions. *Psychology of Addictive Behaviors*, *38*(7), 759–771. https://doi.org/10.1037/adb0001012

2021 Van De Schoot, R., De Bruin, J., Schram, R., Zahedi, P., De Boer, J., Weijdema, F., Kramer, B., Huijts, M., Hoogerwerf, M., Ferdinands, G., Harkema, A., Willemsen, J., **Ma**, **Y.**, Fang, Q., Hindriks, S., Tummers, L., & Oberski, D. L. (2021). An open source machine learning framework for efficient and transparent systematic reviews. *Nature Machine Intelligence*, *3*(2), 125–133. https://doi.org/10.1038/s42256-020-00287-7

Manuscripts in preparation

These studies have been presented at international conferences; presentation details appear below.

Dissertation manuscripts

- *A Stratification Method for Evaluating and Improving the Representativeness of Survey Response.*
- *Leveraging Survey Variables to Estimate Response Propensity in Adaptive Survey Design.*
- *Predicting Survey Participation from LLM-generated Profiles Improves Recruitment Efficiency.*

Collaborative manuscripts

- *The relationship between public understanding of polls and their credibility.*
- *Survey Attitudes and Privacy Concerns: Measurement Invariance across Germany, the Netherlands, and the United States.*
- *Survey Attitudes and Survey Participation—and Vice Versa.*

Presentations

Conference presentations

2026 **Ma, Y.**, & Traugott, M. (2026, February 25–27). *Survey attitudes and survey participation — and vice versa* [Paper presentation]. 8th Current Innovations in Probability-Based Household Internet Panel Research (CIPHER) Conference, Washington, D.C., United States. https://dornsife.usc.edu/cesr/wp-content/uploads/sites/54/2026/03/Ma_CIPHER-2026.pdf Video recording available at https://youtu.be/oC45tHPmJrc?si=XyjWhfdLeymGcQoq.

2025 **Ma, Y.** (2025b, May 14–16). *Replacing nonresponse modeling with active learning on AI-generated textual descriptions of sample units for efficient survey data collection* [Paper presentation]. 80th Annual Conference of the American Association for Public Opinion Research, St. Louis, MO, United States. https://aapor.confex.com/aapor/2025/meetingapp.cgi/Paper/3983

2025 **Ma, Y.** (2025c, July 14–18). *Stratification method for identifying subgroups requiring different survey data collection strategies* [Paper presentation]. 11th Conference of the European Survey Research Association, Utrecht, The Netherlands.

2025 **Ma, Y.**, & Traugott, M. (2025, July 14–18). *Survey attitudes and privacy concerns: Measurement invariance across Germany, the Netherlands, and the United States* [Paper presentation]. 11th Conference of the European Survey Research Association, Utrecht, The Netherlands.

2025 Traugott, M., & **Ma, Y.** (2025a, February 26–28). *Motivated reasoning and acceptance of poll results on a health policy issue* [Paper presentation]. 7th Current Innovations in Probability-Based Household Internet Panel Research (CIPHER) Conference, Washington, D.C., United States. https://dornsife.usc.edu/cesr/wp-content/uploads/sites/54/2025/03/3-Traugott-CIPHER-Presentation-V1.pdf

2025 Traugott, M., & **Ma, Y.** (2025b, May 14–16). *The relationship between public understanding of polls and their credibility* [Paper presentation]. 80th Annual Conference of the American Association for Public Opinion Research, St. Louis, MO, United States. https://aapor.confex.com/aapor/2025/meetingapp.cgi/Paper/3678

2025 Traugott, M., & **Ma, Y.** (2025c, May 12–15). *The role of polls in a democratic society* [Paper presentation]. 78th Annual Conference of the World Association for Public Opinion Research, St. Louis, MO, United States.

2024 **Ma, Y.** (2024b, May 15–17). *Leveraging survey variables to estimate response propensity in Bayesian adaptive survey design: A simulation study* [Paper presentation]. 79th Annual Conference of the American Association for Public Opinion Research, Atlanta, GA, United States. https://aapor.confex.com/aapor/2024/meetingapp.cgi/Paper/2469

2024 **Ma, Y.**, West, B. T., & McCabe, S. E. (2024, May 15–17). *Daily web survey data collection to improve health: Temporal dynamics of cannabis use motives and context, with implications for adaptive interventions* [Paper presentation]. 79th Annual Conference of the American Association for Public Opinion Research, Atlanta, GA, United States. https://aapor.confex.com/aapor/2024/meetingapp.cgi/Paper/2816

2023 **Ma, Y.**, & Miller, P. V. (2023, November 17–18). *Leveraging survey variables to estimate response propensity in Bayesian adaptive survey design: A simulation study* [Paper presentation]. 48th Annual Conference of the Midwest Association for Public Opinion Research, Chicago, IL, United States.

2021 **Ma, Y.**, Mushkudiani, N., & Schouten, B. (2021a, December 1–3). *An optimal stratification approach for addressing nonresponse bias in Bayesian adaptive survey design* [Paper presentation]. 7th International Workshop on Advances in Adaptive and Responsive Survey Design, Online.

2021 **Ma, Y.**, Mushkudiani, N., & Schouten, B. (2021b, July). *An optimal stratification approach for addressing nonresponse bias in Bayesian adaptive survey design* [Paper presentation]. 9th Conference of the European Survey Research Association, Online. https://www.europeansurveyresearch.org/conf2021/uploads/57/3653/67/Ma_ESRA21.pdf

2016 **Ma, Y.** (2016, July 17–18). The change of the sociopolitical attitudes of the middle class in China: 2003–2013 [Paper presentation], In *Social stratification and mobility* [Symposium]. Annual Meeting of the Chinese Sociological Association, Lanzhou, China.

Invited talks

2025 **Ma, Y.** (2025a, October 31). *From Text to Tidy: Reproducible LLM Deployment and Batch Prompt Processing*. Survey Methodology Lab Meeting, Program in Survey and Data Science, University of Michigan, Ann Arbor, MI, United States.

2024 **Ma, Y.** (2024a, August 26). *Introduction to ASReview LAB: AI-Aided Systematic Reviewing*, Department of Early Childhood Education, The Education University of Hong Kong, Hong Kong SAR.

2022 **Ma, Y.** (2022, July 1). *Human-centered AI through reliable graphical user interfaces*. 1st ASReview Users Meeting, Utrecht, The Netherlands.

Teaching

Teaching at the Graduate Level

2024 **University of Michigan, Ann Arbor**, Institute for Social Research
77th Summer Institute in Survey Research Techniques
55th Sampling Program for Survey Statisticians

Teaching Assistant

- **SURVMETH 614: Analysis of Complex Sample Survey Data.** Graded five homework assignments and the final project; monitored project progress and provided formative feedback.
- **SURVMETH 616: Workshop in Survey Sampling Techniques.** Supported in-class exercises, guided discussion, and helped students apply methods from the companion courses.

- **SURVMETH 625: Methods of Survey Sampling.** Graded six homework assignments and the midterm and final examinations; advised students on a multistage sampling project.

Guest Lecturer
SURVMETH 614: Analysis of Complex Sample Survey Data

- Delivered two lectures linking assumptions, implementation in R and Stata, and substantive interpretation: logistic regression [Slides] and multilevel models [Slides].

Public Scholarship

2026 forthcoming — Sage Video *Survey Research Collection*, Sage Publishing

- Invited interview on practical application of survey methods—stages, process, challenges, and lessons learned—produced for university-level teaching

2025 — Podcast Series on Survey Research, European Survey Research Association

- Invited to a special conference episode to discuss my presentations [Video], top research priorities, and overlooked areas in survey research [Video]

Software Development

Maintainer

`asreview` ASReview: Active learning for systematic reviews. An open-source machine learning tool for efficient, transparent, and interactive screening of large textual datasets.

`asreview-wordcloud` Wordclouds: A tool to create a visual impression of the verbal content within a systematic review dataset

Professional Service

Referee

2026 — *Journal of Survey Statistics and Methodology*
Frontiers In Digital Health
Engineering Reports
2025 — *Demography*
2024 — *American Journal of Preventive Medicine*

Professional Affiliations

Memberships

2024 — American Association for Public Opinion Research
2023 — Midwest Association for Public Opinion Research
2021 — European Survey Research Association
2018 — Asian Criminological Society

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